Assignment 4: Employee Payroll

Course: CS101 Computational Thinking | B.Tech 1st Year

Problem Statement

Develop a payroll script. Calculate the gross salary of employees using basic pay, overtime, and allowances. Implement progressive tax deductions based on the gross amount, and output a formatted salary slip.

Algorithm

1. Create a tuple of employee tuples (Employee ID, Basic Pay, Overtime Hours, Incentives).
2. Iterate through employees. Extract details using tuple unpacking.
3. Calculate allowance as a fixed percentage of basic pay.
4. Calculate gross salary: basic + allowance + (overtime * rate) + incentives.
5. Evaluate progressive deductions using `if-elif-else` (tax brackets).
6. Print the organized payroll list.

Python Implementation

```
# (EmpID, Basic, Overtime_Hrs, Incentives)
employees = (
    ("E-101", 40000, 10, 5000),
    ("E-102", 25000, 5, 0),
    ("E-103", 60000, 2, 8000)
)

print("="*45)
print("EMPLOYEE PAYROLL STATEMENT")
print("="*45)

for emp in employees:
    eid, basic, overtime_hrs, incentives = emp

    # Allowances & Overtime
    da_allowance = basic * 0.15 # 15% Dearness Allowance
    overtime_pay = overtime_hrs * 450 # Rs 450 per hr

    gross_salary = basic + da_allowance + overtime_pay + incentives

    # Progressive Tax Deductions
    tax = 0.0
    if gross_salary > 50000:
        tax = (gross_salary - 50000) * 0.20 + (25000 * 0.10)
    elif gross_salary > 25000:
        tax = (gross_salary - 25000) * 0.10

    net_salary = gross_salary - tax

    print("Employee ID : " + eid)
    print("Gross Salary: Rs. " + str(gross_salary))
    print("Tax Deducted: Rs. " + str(tax))
    print("Net Salary  : Rs. " + str(net_salary))
    print("-" * 45)
```